

Review Paper on Micro-Camera Based Airotor

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Abstract

With the Advancement of Technology the present invention related to the Dental Airotor including Micro-camera for capturing and recording the inner view of mouth so that doctors can treat the patients more conveniently. This Airotor also going to have LED light which can incorporated into hand-piece for improvement of visibility and comfort of working. Wireless transmission of capture image or live feeding of treatment Site to destination device so that it will increase the operability for the dentist during treatment. In general this Airotor are used for various purpose such as Oral cleaning, Tooth decay and Plug removal. Due to high speed the friction creates significant heat within bur and with the help of cooling system can be prevent. With this innovative technique those hard to reach and hard to see area are going to be more accessible.

Keywords

Micro-Camera, Airotor (dental hand-piece), LED, Hand-piece Body, Camera Sensor.

Introduction

A dental airotor is a handheld & a mechanical type of device shown in figure 1. which is used to perform a variety of common dental tasks such as removing decay, polishing fillings, performing cosmetic dentistry and altering prostheses. The airotor itself consists of internal mechanical parts which performs a rotational force and provide power to the cutting tip usually a dental burr. This type of apparatus is used to perform various dental operations.



Figure 1. Dental Airtor



Figure 2. LED and a cooling water-spray Dental Airtor

LED and a cooling water-spray shown in figure 2[5].are also incorporated into certain airtor. But even though, in a dental airtor, treatment may take longer amount of time to perform the task due to limitations of viewing angles. So by using micro-camera in a dental airtor, we can reduce the limitations of viewing angles and it will also help to perform treatment with much accuracy, visibility and overall success of the procedure.

Literature Review

The airtor or handpiece is most commonly used in dentistry to perform majority of dental procedure. Initially a slow speed handpiece resulted in development of heat, pressure which mostly caused death of the pulp. Therefore Clinicians started looking for high speed handpiece with the aim to avoid injury to pulp and speed up the procedure. With the increase in speed, heat generated also increase this began the search for suitable coolant. This led to the incorporation of air and water spray as a coolant. But because of increased speed and water coolant, the area of cutting of tooth structure was blocked by water spray. Hence ,search began in the direction to enhance the vision.

Fiber optics light was introduced to enhance the visibility. Most recent invention in optics is a LED.

Development in the handpiece impacts all aspect of dental treatment. Though dental handpiece is the most fundamental instrument used by dentist, but it is also the instrument most likely to cause tissue damage. Add-on of micro camera will widely improve the accuracy, visibility, precision, will help prevent dental mishaps, preserve the integrity of tooth. With the ever changing technology and tools,dentists choices are greatly influenced by their own comfort and facility and incorporation of microcamera can drastically improve the accuracy of dental procedure.

Dental hand-piece used in [1], the dentist perform operations as well as simultaneously can also watch live clear color internal view of patient's mouth on display device by using camera fixed inside hand-piece. Camera

system consists of color image sensor (CMOS) used to provide color image signal that is transmitting to display on monitor and condensing lens unit with focus range of 3-50 mm for wide clear color image of treatment site, it means high practical value can be provided. The number of light-emitting diode (LED's) are arranged around the camera unit to provide better visibility of treatment site. Dental hand-piece also helps in preventing visibility from fogging effect (air curtain effect) using air coming out from entrance of anti-fogging passage of light receiving port created by spray or water droplets during cutting.

In [2], system provides a removable camera attachment mounting onto the drilling end for a low-speed dental hand-piece and a cavity viewing system. Hand-piece also provides a silicone-encased camera board with a lens that magnifies the cavity preparation site and an LED that illuminates the site. Since Low-speed Dental Hand-piece does not use water, the images produced by the camera are clear. Camera attachment is located external to the hand-piece, it can be easily removed, cleaned, sterilized, and replace. This System improves a dentist's ability to assess the complete removal of decay.

In [3] wirelessly transmit camera images to various destination devices such that it will increase operability for a dentist or during a treatment and achieve improved visibility of an affected area and also patient understanding of the details of the treatment. A driving voltage is supplied from the power supply circuit unit of the image processing wireless transmission unit to the camera through the coupling connector unit, the cylindrical connector unit and the connection wiring unit and the image signal of the target portion imaged by the camera is connected to the camera into the color image generation unit and the cylindrical unit connector unit, generating the target part of color image information and is configured to wirelessly transmit from the transmission transmitter to the destination device.

Current Limitations :

1. Only limited view of treatment site shows using conventional mirror used by the dentist.
2. No cleaning system for the Micro Camera attached to Hand-piece.
3. Dental Hand-piece produces characteristics noise that is harmful for both Patient & Dentist.
4. Risk of human error while treatment.
5. Less Accuracy.

Proposed Solution :

1. The proposed solution for the above current limitations is that to design a Micro Camera based Airtor, where dental hand-piece combined with built-in micro camera.

2. Micro-camera based Airotor device will be use to real time photographiing of the treatment site and display the capture image information on the screen.

Objectives:

1. To get better & clear view of the treatment site of the oral cavity.
2. To take real time images of inner & outer surface of cavity to analysis & planning for the operation.
3. Noise levels of the Airotor will be reduced.
4. Dental treatment analysis and sharing of reports may get easier.

Scope of Work:

1. The dental Airotor hand-piece scope is that it consists of micro camera device with the ability to take visual information from the operation site of cavity and project it onto screen for the dentist to view.
2. This hand piece is a hand-held.
3. It includes different dental procedures like removing decay, polishing fillings, removing cosmetic dentistry and altering prosthesis

Expected Outcomes-:

1. The expected outcomes of Micro-camera based Airotor is to displays the real time video shooting of treatment site of the oral cavity on the screen.
2. This helps the dentist to teach the internship students or interns real time inside view of the oral cavity on the screen.

Conclusion

In using the present invention, the dentist will place the monitor in a convenient viewing location and turn on the master switch and the LED. Using the micro camera based airotor, on which the camera viewing attachment is mounted, the dentist will activate the excavating bur in order to remove decay from a patient's cavity. As the decay is removed from the cavity, the dentist will be able to see a magnified, real-time image and recording for future reference of their work. The Dentist will be better able to see microscopic nerve exposures allowing for a better treatment decision and diagnosis. If a tooth requires a root canal, the excavation bur can be used to access the pulpal chamber and then can be replaced with a root canal file for enhanced viewing of the canals during

canal instrumentation.

Although there are many details in the description, it should not be construed as limiting the scope of the invention, but only as examples illustrating the current preference of this invention.

References

1. Takeshi Hasegawa, Michizo Yamanaka, "Hand-piece with built-in camera and dental treatment method", E.P. Patent 2891467 A1, July 08, 2015.
2. John Jesus Rodriquez, Jr, " Camera Viewing Attachment & Display System for Low Speed Dental Hand-piece", U.S. Patent 10306124 B1, May 28, 2019. .
3. John S Seney, "Dental bur cooling system", U.S. Patent 5096421A, March 17, 1992
4. Johann Hain, Ulf Stein, "Dental hand-piece with image transfer device" A.T. Patent 396740B, November 25, 1993.
5. <https://www.indiamart.com/proddetail/waldent-led-airotor-handpiece-22487627297.html>